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## Ultrasonic diagnostic apparatus and ultrasonic probe


### Abstract

An ultrasonic diagnostic apparatus according to the present disclosure includes: an ultrasonic probe connected to an apparatus body, comprising a battery that supplies power to components of the ultrasonic probe and a cooler that cools inside the ultrasonic probe; and a drive controller that drives the cooler based on a remaining level of the battery and an internal temperature of the ultrasonic probe.



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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2022-127414, filed on Aug. 9, 2022, the entire contents of which are incorporated herein by reference.
FIELD
(2) The embodiments disclosed in the present specification and drawings relate to an ultrasonic diagnostic apparatus and an ultrasonic probe.
BACKGROUND
(3) In today's medical fields, ultrasonic diagnostic apparatuses are widely used to examine inside a subject by using ultrasonic waves generated with an ultrasonic probe. To examine inside the subject with the ultrasonic diagnostic apparatus, a subject contact of an ultrasonic probe having an ultrasonic transducer that transmits/receives ultrasonic waves should be brought into contact with the subject. For this reason, the international standard IEC60601-2-37 defines a temperature standard of the subject contact, and the temperature of the subject contact of the ultrasonic probe should be prevented from rising above a predetermined temperature.
(4) Especially, in a wireless ultrasonic probe that connects an apparatus body with the ultrasonic probe of the ultrasonic diagnostic apparatus with wireless communication, a processing circuitry that performs signal processing etc. and a battery that supplies power to components of the ultrasonic probe are housed inside a housing case of the ultrasonic probe. Therefore, a usable time of the ultrasonic probe is limited by both a rising temperature of the ultrasonic probe due to heat generated by the processing circuitry etc. and a remaining level of the battery.

(5) Here, the ultrasonic probe, in which the cooling unit is driven when an internal temperature exceeds a predetermined threshold, is proposed to suppress the rising temperature of the ultrasonic probe. Also proposed is the ultrasonic probe that shifts to a power savings mode when the remaining level of the battery falls below the predetermined threshold. However, with such ultrasonic probe, there is a concern of affecting the diagnosis using the ultrasonic diagnostic apparatus, since a performance of the ultrasonic probe may change regardless of an operator's intention. Moreover, there is a problem of being unable to secure a sufficient usable time of the ultrasonic probe, because the timing when the power of the battery depletes and the timing when the ultrasonic probe becomes unusable due to the rising temperature do not match.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view schematically illustrating an exterior of the ultrasonic diagnostic apparatus according to a first embodiment.
(2) FIG. **2** is a block diagram illustrating an exemplary electrical configuration of the ultrasonic diagnostic apparatus according to the first embodiment.
(3) FIG. **3** is a description diagram describing an exemplary internal layout of the ultrasonic diagnostic apparatus according to the first embodiment.
(4) FIG. **4** is a diagram illustrating a graph showing a relationship between a remaining level of a battery of an ultrasonic probe and a usable time of the ultrasonic probe which the ultrasonic diagnostic apparatus according to the first embodiment includes.
(5) FIG. **5** is a diagram illustrating a graph showing a relationship between an internal temperature of the ultrasonic probe and the usable time of the ultrasonic probe which the ultrasonic diagnostic apparatus according to the first embodiment includes.
(6) FIG. **6** is a diagram illustrating a graph showing a change of the usable time of the ultrasonic probe when a cooler is driven at some time.
(7) FIG. **7** is a diagram illustrating a flowchart describing contents of a cooling control process executed in the ultrasonic diagnostic apparatus according to the first embodiment.
(8) FIG. **8** is a diagram illustrating the internal layout of the ultrasonic probe describing a modified configuration of the ultrasonic probe.
(9) FIG. **9** is a block diagram illustrating the exemplary electrical configuration of the ultrasonic diagnostic apparatus according to a second embodiment.
DETAILED DESCRIPTION
(10) With reference to the drawings below, embodiments of an ultrasonic diagnostic apparatus and an ultrasonic probe will be described. In the description below, note that same reference signs are given for components substantially identical in terms of configuration and function and duplicate description will be given only when necessary.
First Embodiment
(11) FIG. **1** is a schematical diagram illustrating an exterior of the ultrasonic diagnostic apparatus according to a first embodiment. As shown in FIG. **1**, the ultrasonic diagnostic apparatus **1** according to the present embodiment includes the ultrasonic probe **10**, an apparatus body **30**, a display **50**, and an input interface **70**. Note that in the example shown in FIG. **1**, the display **50** and the input interface **70** are configured separately from the apparatus body **70**, but the apparatus body **30**, the display **50**, and the input interface **70** may be configured as one. An example where the apparatus body **30**, the display **50**, and the input interface **70** are configured as one, for instance, includes an information processing device such as a tablet terminal or a workstation.
(12) The ultrasonic probe **10** refers to a device connected to the apparatus body **30**, which transmits an ultrasonic signal to a subject and receives a reflected wave signal based on the transmitted

ultrasonic signal reflected within the subject. The ultrasonic probe **10**, for instance, includes an 1D array probe which scans a 2-dimension area within the subject, or a mechanical 4D probe or a 2D array probe which scans a 3-dimension area within the subject. Although FIG. **1** illustrates an example where the ultrasonic probe **10** is connected to the apparatus body **30** with wireless communication, the ultrasonic probe **10** may be arranged to be connected to the apparatus body **30** with wired communication such as those via cables. Note that in the description below, the ultrasonic probe **10** according to the present embodiment will be described with an example where the ultrasonic probe **10** is arranged to be connected to the apparatus body **30** with wireless communication.

(13) The apparatus body **30** refers to a device that generates ultrasonic images based on received signals received from the ultrasonic probe **10**. In the example shown in FIG. **1**, the ultrasonic probe **10**, the display **50**, and the input interface **70** are connected to the apparatus body **30** according to the present embodiment.

(14) The display **50** refers to a display device that displays various ultrasonic images or various settings etc. For instance, the display **50** may display ultrasonic images generated by the apparatus body **30** or various settings performed by an operator. In the present embodiment, the display **50**, for instance, is configured by a liquid crystal display or a Cathode Ray Tube (CRT) display etc. If the display **50** is configured by a display device that allows operation and input, the display **50** also displays a Graphical User Interface (GUI) etc., to receive various operations from the operator.

(15) The input interface **70** refers to an input device to perform various settings etc., and is realized by, for instance, a trackball, a switch button, a mouse, a keyboard, a touchpad that performs input operations by touching an operating screen, a touch monitor where a display screen is combined with the touchpad, a non-touch input circuitry using a photosensor, or an audio input circuitry, etc. The input interface **70** connected to the processing circuitry of the apparatus body **30** which will be described later, converts the input operation received from the operator into electrical signal and outputs to the processing circuitry of the apparatus body **30**. Note that the input interface **70** according to the present disclosure is not limited to those having physical components for operation such as the mouse or keyboard. For instance, the processing circuitry of the electrical signal that receives electrical signals corresponding to input operations from an external input device provided separately from the apparatus and outputs the electrical signal to the processing circuitry of the apparatus body **30**, is also included as an example of the input interface **70**.

(16) Next, referring to FIG. **2**, an exemplary configuration of the ultrasonic diagnostic apparatus **1** according to the present embodiment will be described. FIG. **2** is a block diagram illustrating an exemplary electrical configuration of the ultrasonic diagnostic apparatus **1** according to the first embodiment. As also described in FIG. **1**, the ultrasonic diagnostic apparatus **1** according to the present embodiment shown in FIG. **2** includes the ultrasonic probe **10**, the apparatus body **30**, the display **50**, and the input interface **70**.

(17) In further detail, the ultrasonic probe **10** which the ultrasonic diagnostic apparatus **1** according to the present embodiment includes, includes a subject contact **101**, a transceiver circuitry **103**, a processing circuitry **105**, a cooler **107**, a battery **109**, a communication circuitry **111**, and a temperature sensor **113**. In the present embodiment, these components are stored in a single housing case. On the other hand, the apparatus body **30** which the ultrasonic diagnostic apparatus **1** according to the present embodiment includes, includes a processing circuitry **301**, a memory circuitry **303**, and a communication circuitry **305**.

(18) First, describing about the configuration of the ultrasonic probe **10**, the subject contact **101**, for instance, transmits ultrasonic waves to the subject P via a subject contact surface that contacts with the subject P on a transceiver side of the ultrasonic waves and receives the reflected waves of the transmitted ultrasonic waves reflected within subject P. Thus, the subject contact **101** has an ultrasonic transducer that transmits/receives ultrasonic waves.

(19) The ultrasonic transducer, for instance, refers to a piezoelectric oscillator made of piezoelectric

ceramic. The ultrasonic transducer is provided as a plurality of arrays on a tip of the subject contact **101**, transmits ultrasonic waves based on a drive signal supplied from the transceiver circuitry **103**, receives reflected waves from inside of the subject P, and converts the reflected waves to electrical signals.

(20) The transceiver circuitry **103** refers to a processor having a transmitter circuitry and a receiver circuitry that controls a transmission directivity and a receiving directivity in transmitting/receiving the ultrasonic waves. Note that in FIG. **2**, an example was shown where the transceiver circuitry **103** is provided in the ultrasonic probe **10**, but the transceiver circuitry **103** may be provided in both the ultrasonic probe **10** and the apparatus body **30**.

(21) The transmitter circuitry includes a pulse generator, a transmission delay circuitry, and a pulsar circuitry etc., and supplies the drive signal to the ultrasonic transducer. The pulse generator repeatedly generates a rate pulse to form the transmitting ultrasonic signal at a preset rate frequency. The transmission delay circuitry applies a delay time for every piezoelectric oscillation necessary to converge the ultrasonic wave generated from the ultrasonic transducer into a beam and determine the transmission directivity, to each rate pulse generated by the pulse generator. Further, the pulsar circuitry applies a drive pulse to the ultrasonic transducer at timings based on the rate pulse. The transmission delay circuitry arbitrarily adjusts a transmission direction of the ultrasonic wave transmitted from an ultrasonic transducer surface by modifying the delay time applied to each rate pulse.

(22) The receiver circuitry includes an amplifier circuitry, A/D converter, an adder etc., receives reflected wave signals based on the reflected waves received by the ultrasonic transducer, and generates the received signal (echo signal) by performing various processing to the received wave signals. The amplifier circuitry amplifies reflected wave signals for each channel and performs gain correction processing. The A/D converter A/D converts the gain-corrected reflected wave signals and applies the delay time necessary to determine the receiving directivity to the digital data. The adder generates the received signals by performing an adding process of the reflected wave signals processed by the A/D converter. A reflective component from a direction in response to the receiving directivity of the reflected wave signals, is enhanced by the adding process of the adder.

(23) The processing circuitry **105**, for instance, refers to a control circuitry that performs an overall control of the ultrasonic probe **10**, and for instance is configured by processors such as a CPU or GPU. The processing circuitry **105** according to the present embodiment controls the transceiver circuitry **103**, the cooler **107**, or the battery **109**.

(24) The cooler **107** refers to a cooling device that cools inside the ultrasonic probe **10**. The cooler **107** may be configured by an air-cooling device using a fan etc., a liquid cooling device where a heat conducting fluid flows inside a pipe arranged within the ultrasonic probe **10**, or an electrical cooling device such as a Piezo element or a Peltier element. In the present embodiment, the cooler **107** may turn its drive ON or OFF. Power is supplied from the battery **109** when driving the cooler **107**. In other words, the remaining level of the battery **109** decreases by driving the cooler **107**.

(25) The battery **109** supplies power to each component of the ultrasonic probe **10**. In the present embodiment, the battery **109** is configured by, for instance, rechargeable batteries such as lithium-ion batteries or nickel-metal hydride batteries. The battery **109**, for instance, supplies power to the transceiver circuitry **103**, the processing circuitry **105**, the cooler **107**, or the communication circuitry **111** etc. Note that the battery **109** may use non-rechargeable batteries as well as rechargeable batteries.

(26) The communication circuitry **111** implements various protocols for wireless communications, realizing communication with the apparatus body **30** with wireless communication. The communication circuitry **111** according to the present embodiment, for instance, transmits the received signal generated by the receiver circuitry of the transceiver circuitry **103** to the apparatus body **30** with wireless communication. Note that in the example of FIG. **1** according to the present embodiment, the ultrasonic probe **10** and the apparatus body **30** communicates with wireless

communication, but the communication between the ultrasonic probe **10** and the apparatus body **30** is not limited to wireless communication. For instance, if the communication between the ultrasonic probe **10** and the apparatus body **30** is done with wired communication, the communication circuitry **111** may implement various protocols for wired communication, realizing communication between the ultrasonic probe **10** and the apparatus body **30** with wired communication via such as cables.

(27) The temperature sensor **113** refers to a sensor that detects an internal temperature of the ultrasonic probe **10**. The temperature sensor **113** may be configured by, for instance, sensors such as a thermocouple, a resistance temperature detector, a thermistor, or an IC temperature sensor. Especially in the present embodiment, the temperature sensor **113** detects the temperature of the subject contact **101** or its vicinity, and outputs the detected temperature to the processing circuitry **105** as the internal temperature of the ultrasonic probe **10**.

(28) FIG. **3** is a diagram that describes an exemplary internal layout of the ultrasonic probe **10** which the ultrasonic diagnostic apparatus **1** according to the present embodiment includes. As shown in FIG. **3**, the ultrasonic probe **10** according to the present embodiment includes a thermal conductor **120**. The thermal conductor **120**, for instance, is configured by heat transfer members consisting of graphite or metals with high thermal conductivity such as aluminum or copper. Also, the thermal conductor **120** is arranged to be cooled by the cooler **107**.

(29) In the present embodiment, the thermal conductor **120** connects the subject contact **101** and the battery **109** so that heat may be transferred to each other between at least the subject contact **101** and the battery **109**. By this, the subject contact **101** and the battery **109** may be efficiently cooled to suppress the temperature rise of these.

(30) Likewise, the thermal conductor **120** may be formed and arranged so that heat is also transferred to each other between the transceiver circuitry **130**, the processing circuitry **105**, and the communication circuitry **111**, in addition to the subject contact **101** and the battery **109**. By this, the transceiver circuitry **103**, the processing circuitry **105**, and the communication circuitry **111** may also be efficiently cooled by the cooler **107**.

(31) Returning to FIG. **2** again, the processing circuitry **301**, the memory circuitry **303**, and the communication circuitry **305** that configures the apparatus body **30** will be described.

(32) The processing circuitry **301** refers to a control circuitry that performs an overall control of the ultrasonic diagnostic apparatus **1**. Likewise, the processing circuitry **301** also refers to an arithmetic circuitry that performs various calculations, and for instance, is configured by processors such as CPU or GPU. The processing circuitry **301** according to the present embodiment, for instance, generates ultrasonic image data based on received signals via the communication circuitry **305** or displays the generated ultrasonic image data on the display **50**.

(33) Thus, the processing circuitry **301** according to the present embodiment has a B-mode processing function **3011**, a Doppler processing function **3012**, an image generation function **3013**, a display control function **3014**, a remaining battery acquiring function **3015**, a first calculation function **3016**, an internal temperature acquiring function **3017**, a second calculation function **3018**, and a drive determination function **3019**. The B-mode processing function **3011** is equivalent to a B-mode processor according to the present embodiment; the Doppler processing function **3012** is equivalent to a Doppler processor according to the present embodiment; the image generation function **3013** is equivalent to an image generator according to the present embodiment; and the display control function **3014** is equivalent to a display controller according to the present embodiment. Also, the remaining battery acquiring function **3015** is equivalent to a remaining battery acquirer according to the present embodiment; the first calculation function **3016** is equivalent to a first calculator according to the present embodiment; the internal temperature acquiring function **3017** is equivalent to an internal temperature acquirer according to the present embodiment; the second calculation function **3018** is equivalent to a second calculator according to the present embodiment; and the drive determination function **3019** is equivalent to a drive

determiner according to the present embodiment. Also, the processing circuitry **301** is equivalent to a drive controller according to the present embodiment.

(34) In the embodiment shown in FIG. **2**, each processing function performed in the B-mode processing function **3011**, the Doppler processing function **3012**, the image generation function **3013**, the display control function **3014**, the remaining battery acquiring function **3015**, the first calculation function **3016**, the internal temperature acquiring function **3017**, the second calculation function **3018**, and the drive determination function **3019** is stored in the memory circuitry **303** in a form of computer executable program. The processing function **301** refers to a processor that realizes functions corresponding to each program by reading and executing the program from the memory circuitry **303**. In other words, the processing circuitry **301** that has read each program has each function shown in the processing circuitry **301** in FIG. **2**. Note that it was described in FIG. **2** to realize the B-mode processing function **3011**, the Doppler processing function **3012**, the image generation function **3013**, the display control function **3014**, the remaining battery acquiring function **3015**, the first calculation function **3016**, the internal temperature acquiring function **3017**, the second calculation function **3018**, and the drive determination function **3019** with a single processing circuitry **301**, but these functions may be realized by combining a plurality of independent processors to arrange the processing circuitry **301** and let each processor execute the program.

(35) The B-mode processing function **3011** refers to a function that generates B-mode data based on received signals received from the ultrasonic probe **10**. The processing circuitry **301** performs, for instance, an envelope detection or a logarithmic compression to the received signal from the ultrasonic probe **10** with the B-mode processing function **3011** and generates B-mode data where a signal intensity is represented by brightness of luminosity.

(36) The Doppler processing function **3012** refers to a function that generates Doppler data such as a blood flow velocity, a blood flow dispersion, or a blood flow power based on received signals received from the ultrasonic probe **10**. The Doppler processing function **3012** generates Doppler data which extracts blood flow information based on a Doppler effect of a moving body within a region of interest (ROI) set to a scanning region by frequency analyzing the received signals received from the ultrasonic probe **10**.

(37) The image generation function **3013** refers to a function that generates the ultrasonic image data based on received signals received from the ultrasonic probe **10**. Specifically, the image generation function **3013** refers to a function that generates various ultrasonic image data based on B-mode data generated by the B-mode processing function **3011** and/or Doppler data generated by the Doppler processing function **3012**. For instance, the image generation function **3013** generates a B-mode image data representing the intensity of the reflected waves as luminance, from the B-mode data generated by the B-mode processing function **3011**. Also, for instance, the image generation function **3013** generates an average velocity image, a dispersion image, a power image, or a combined image of these as a Doppler image data, from the Doppler data generated by the Doppler processing function **3012**.

(38) The display control function **3014** refers to a function that displays various ultrasonic image data generated by the image generation function **3013** on the display **50**. Specifically, for instance, the display control function **3014** controls the display of the B-mode image data, the Doppler image data, or the ultrasonic image data including both, generated by the image generation function **3013**, on the display **50**.

(39) The remaining battery acquiring function **3015** acquires the remaining battery level of the ultrasonic probe **10**. Specifically, the remaining battery acquiring function **3015** acquires the remaining level of the battery **109** via the communication circuitry **305** and the communication circuitry **111**. In the present embodiment, for instance, the remaining level of the battery **109** may be managed in the battery **109** itself or be managed by the processing circuitry **105**. Also, the remaining level of the battery **109**, for instance, may be represented as a percentage of the

remaining level at that time with a fully charged state as 100% and a fully discharged state as 0%.
(40) Also, the remaining battery acquiring function **3015** records the acquired remaining level of the battery **109** over time as a detection result. The recording of the detection result over time may be held by the remaining battery acquiring function **3015** or stored and held in the memory circuitry **303**.
(41) The first calculation function **3016** calculates a first usable time, which is the remaining usage time of the ultrasonic probe **10** corresponding to the remaining level, based on the remaining level of the battery **109** acquired by the remaining battery acquiring function **3015**.
(42) The internal temperature acquiring function **3017** acquires the internal temperature of the ultrasonic probe **10**. Specifically, the internal temperature **3017** acquires the internal temperature of the ultrasonic probe **10** via the communication circuitry **305** and the communication circuitry **111**. As described above, the internal temperature of the ultrasonic probe **10** is regularly detected and output to the processing circuitry **105** by the temperature sensor **113**. Thus, the internal temperature acquiring function **3017** acquires the temperature detected by the temperature sensor **113** as the internal temperature of the ultrasonic probe **10**.
(43) Also, the internal temperature acquiring function **3017** records the acquired internal temperature of the ultrasonic probe **10** as the detection result over time. The record of the detection result over time may be held by the internal temperature acquiring function **3017** or stored and held in the memory circuitry **303**.
(44) The second calculation function **3018** calculates a second usable time, which is the remaining usage time of the ultrasonic probe **10** corresponding to the internal temperature of the ultrasonic probe **10**, based on the internal temperature acquired by the internal temperature acquiring function **3017**.
(45) The drive determination function **3019** determines whether to drive the cooler **107** based on the first usable time calculated by the first calculation function **3016** and the second usable time calculated by the second calculation function **3018**. In other words, the drive determination function **3019** determines whether to drive an inactive cooler **107**. Further, the drive determination function **3019** determines whether to stop the cooler **107** based on the first usable time calculated by the first calculation function **3016** and the second usable time calculated by the second calculation function **3018**. In other words, the drive determination function **3019** determines whether to stop an active cooler **107**.
(46) The memory circuitry **303** may be realized by semiconductor elements such as a random access memory (RAM) or a flash memory etc., a hard disk, or an optical disk etc. In the present embodiment, for instance, the memory circuitry **303** stores the program executed in the circuitry included in the ultrasonic probe **10** or the apparatus body **30**.
(47) The communication circuitry **305** implements various protocols for telecommunication in response to a form of a network. The communication circuitry **305** realizes communication with other apparatus via the network following these various protocols. For instance, the communication circuitry **305** according to the present embodiment receives the received signals generated by the receiver circuitry of the transceiver circuitry **103** of the ultrasonic probe **10**, transmits a control program of the ultrasonic probe **10** to the ultrasonic probe **10**, or transmits ultrasonic images etc., generated by the apparatus body **30** to other devices. As described above, in the present embodiment, the communication circuitry **305** and the communication circuitry **111** communicates with wireless communication but may communicate with wired communication as well.
(48) FIG. **4** is a diagram illustrating a graph showing a relationship between the remaining level of the battery **109** of the ultrasonic probe **10** and the usable time of the ultrasonic probe **10** which the ultrasonic diagnostic apparatus **1** according to the present embodiment includes. Solid lines in FIG. **4** denote the detection result of the remaining level of the battery **109** so far. With a premise that this change continues, the remaining battery level is estimated to change from now on as shown with a dotted line. However, the ultrasonic probe **10** cannot be properly used when the remaining

battery level falls below a first threshold, which is a predetermined remaining level. Thus, the first calculation function **3016** calculates the time until the remaining level of the battery **109** falls below the first threshold as the first usable time, which is the remaining usage time of the ultrasonic probe **10**. In other words, the first usable time denotes the time from a current time T**1** until the remaining level of the battery **109** falls below the first threshold.
(49) FIG. **5** is a diagram illustrating a graph showing a relationship between the internal temperature of the ultrasonic probe **10** and the usable time of the ultrasonic probe **10** which the ultrasonic diagnostic apparatus **1** according to the present embodiment includes. The solid lines in FIG. **5** denotes the detection result of the internal temperature of the ultrasonic probe **10** until now. With the premise that this change continues, the internal temperature is estimated to change from now on as shown with the dotted line. However, considering effects to a patient, the ultrasonic probe **10** cannot be used when the internal temperature rises above a second threshold, which is a predetermined temperature. Thus, the second calculation function **3018** calculates the time until the internal temperature of the ultrasonic probe **10** rises above the second threshold as the second usable time, which is the remaining usage time of the ultrasonic probe **10**. In other words, the second usable time denotes the time from a current time T**2** until the internal temperature of the ultrasonic probe **10** rises above the second threshold.
(50) Here, the second usable time which depends on the internal temperature of the ultrasonic probe **10** extends when the cooler **107** is driven to suppress the internal temperature rise of the ultrasonic probe **10**. However, the power of the battery **109** will be consumed when the cooler **107** is driven. Thus, the first usable time depending on the remaining level of the battery **109** shortens. As such, the relationship between the first usable time and second usable time trade-offs.
(51) FIG. **6** is a diagram illustrating a graph showing a change of the usable time of the ultrasonic probe **10** when the cooler **107** is driven at some time T**3**. In FIG. **6**, a bold line represents the change of the first usable time, which is the remaining usage time based on the remaining level of the battery **109**, and a thin line represents the change of the second usable time, which is the remaining usage time based on the internal temperature of the ultrasonic probe **10**. In both the bold and thin lines, the solid line represents an estimate when driving the cooler **107** and an actual value, which is the detection result so far, and the dotted line represents an estimate when the cooler **107** was not driven.
(52) As may be known from the change of the solid and dotted bold lines after time T**3** in the graph of FIG. **6**, an inclination of the solid line becomes steeper than that of the dotted line when the cooler **107** is driven, and the first usable time, which is the remaining usage time based on the remaining level of the battery **109**, shortens since the power of the battery **109** is consumed. On the other hand, as may be known from the change of the solid and dotted lines related to the thin line after time T**3** in the graph of FIG. **6**, an inclination of the solid line becomes lenient than that of the dotted line when the cooler **107** is driven, and the second usable time, which is the remaining usage time based on the internal temperature of the ultrasonic probe **10**, extends.
(53) Here, to allow the operator to use the ultrasonic probe **10** as long as possible, it is preferable to have the difference between the first usable time, which is the remaining usage time based on the remaining level of the battery **109**, and the second usable time, which is the remaining usage time based on the internal temperature of the ultrasonic probe **10**, as small as possible. In other words, in the graph of FIG. **6**, it is preferable to have the solid part of the bold line showing the first usable time and the solid part of the thin line showing the second usable time to match as possible.
(54) There, in the ultrasonic diagnostic apparatus **1** according to the present embodiment, the drive of the cooler **107** is controlled based on the remaining level of the battery **109** and the internal temperature of the ultrasonic probe **10**. Specifically, the drive of the cooler **107** is controlled based on the first usable time, which is the remaining usage time based on the remaining level of the battery **109**, and the second usable time, which is the remaining usage time based on the internal temperature of the ultrasonic probe **10**; and the difference between the first usable time, which is

the remaining usage time corresponding to the remaining level of the battery **109**, and the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**, is made as small as possible.

(55) To control the drive of the cooler **107** as such, the ultrasonic diagnostic apparatus **1** according to the present embodiment executes a cooling control process as shown in FIG. **7**. In other words, FIG. **7** is a diagram illustrating a flowchart describing contents of the cooling control process executed in the ultrasonic diagnostic apparatus **1** according to the present embodiment. In the present embodiment, the cooling control process shown in FIG. **7**, for instance, is executed in the processing circuitry **301** in the apparatus body **30** of the ultrasonic diagnostic apparatus **1**. Likewise, the cooling control process refers to a process regularly executed when the ultrasonic diagnostic apparatus **1** is in operation.

(56) As shown in FIG. **7**, in the cooling control process, the ultrasonic diagnostic apparatus **1** first acquires the remaining level of the battery **109** (Step S**11**). Specifically, the remaining battery acquiring function **3015** in the processing circuitry **301** of the ultrasonic diagnostic apparatus **1** acquires the remaining level of the battery **109** from the ultrasonic probe **10** at that time.

(57) Next, the ultrasonic diagnostic apparatus **1** calculates the first usable time, which is the remaining usage time of the ultrasonic probe **10** corresponding to the remaining level of the battery **109** based on the remaining level of the battery **109** acquired at Step S**11** (Step S**13**). Specifically, the first calculation function **3016** in the processing circuitry **301** of the ultrasonic diagnostic apparatus **1** calculates the first usable time based on the remaining level of the battery **109** acquired by the remaining battery acquiring function **3015**. The first calculation function **3016**, for instance, as shown in FIG. **4**, estimates the change of the remaining battery level thereafter from the detection result of the remaining level of the battery **109** so far and calculates the time until falling below the first threshold, which is the remaining operable battery level, as the first usable time.

(58) Next, as shown in FIG. **7**, the ultrasonic diagnostic apparatus **1** acquires the internal temperature of the ultrasonic probe **10** (Step S**15**). Specifically, the internal temperature acquiring function **3017** in the processing circuitry **301** of the ultrasonic diagnostic apparatus **1** acquires the internal temperature of the ultrasonic probe **10** at that time from the ultrasonic probe **10**.

(59) Next, the ultrasonic diagnostic apparatus **1** calculates the second usable time, which is the remaining usage corresponding to the internal temperature of the ultrasonic probe **10** based on the internal temperature of the ultrasonic probe **10** acquired at Step S**15** (Step S**17**). Specifically, the second calculation function **3018** in the processing circuitry **301** of the ultrasonic diagnostic apparatus **1** calculates the second usable time based on the internal temperature of the ultrasonic probe **10** acquired by the internal temperature acquiring function **3017**. The second calculation function **3018**, for instance, as shown in FIG. **5**, estimates the change of the internal temperature thereafter from the detection result of the internal temperature of the ultrasonic probe **10** so far and calculates the time until rising above the second threshold, which is an operable temperature, as the second usable time.

(60) Next, as shown in FIG. **7**, the ultrasonic diagnostic apparatus **1** determines whether to drive or stop the cooler **107** (Step S**19**). Specifically, the drive determination function **3019** in the processing circuitry **301** of the ultrasonic diagnostic apparatus **1** determines whether to drive the inactive cooler **107** or stop the active cooler **107** based on the first usable time calculated by the first calculation function **3016** and the second usable time calculated by the second calculation function **3018**.

(61) For instance, when the cooler **107** is inactive, the drive determination function **3019** determines to drive the cooler **107** when the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, is longer than the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**. By such, although the power of the battery **109** is used to drive the cooler **107**, the internal temperature rise of the ultrasonic probe **10** is suppressed, decreasing the difference between the

first usable time and the second usable time, and changing in a direction where the first usable time and the second usable time match.

(62) On the other hand, when the cooler **107** is active, the drive determination function **3019** determines to stop the cooler **107** when the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, is shorter than the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**. By such, although the internal temperature rise of the ultrasonic probe **10** is not suppressed, the power of the battery **109** to drive the cooler **107** is not used, decreasing the difference between the first usable time and the second usable time and changing in a direction where the first usable time and the second usable time match.

(63) When driving or stopping the cooler **107**, the drive determination function **3019** may have a grace period when switching the drive and stop. For instance, when the drive determination function **3019** drives the cooler **107**, the cooler **107** is continuously driven for 5 minutes thereafter. In contrast, when the drive determination function **3019** stops the cooler **107**, the cooler **107** is not driven for 5 minutes thereafter. By such, it becomes possible to prevent the cooler **107** from repeating the drive and stop in a short period of time.

(64) Next, as shown in FIG. **7**, when determining to drive or stop the cooler **107** at Step S**19** (Step S**19**: Yes), the ultrasonic diagnostic apparatus **1** drives the inactive cooler **107** or stops the active cooler **107** (Step S**21**). Specifically, the drive determination function **3019** in the processing circuitry **301** of the ultrasonic diagnostic apparatus **1** sends a command to the processing circuitry **105** of the ultrasonic probe **10** via the communication circuitry **305** and the communication circuitry **111**, commanding to drive or stop the cooler **107**.

(65) After the drive determination function **3019** commands to drive or stop the cooler **107** at Step S**21** and when the drive determination function **3019** determines that driving or stopping the cooler **107** is unnecessary at Step S**19** described above (Step S**19**: No), the drive determination function **3019** returns to Step S**11** described above and repeats the process from Step S**11**.

(66) Note that the cooler **107** not only performs the simple control of drive and stop but also may adjust an intensity of a cooling function. For instance, the cooler **107** may have three levels (strong, medium, weak) as the intensity of the cooling function. In such case, the drive determination function **3019** may drive the cooler **107** with cooling functions of different intensity based on how great the difference between the first usable time and the second usable time is.

(67) For instance, the difference between the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, and the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**, may be divided into three by a first difference and a second difference. Here, the second difference is greater than the first difference.

(68) Then, if the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, is longer than the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**, and if the difference between the first usable time and the second usable time is less than the first difference, the drive determination function **3019** drives the cooler **107** with the cooling function of weak level.

(69) On the other hand, if the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, is longer than the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**, and if the difference between the first usable time and the second usable time is greater than or equal to the first difference and less than the second difference, the drive determination function **3019** drives the cooler **107** with the cooling function of medium level.

(70) Further, if the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, is longer than the second usable time, which is the remaining

usage time corresponding to the internal temperature of the ultrasonic probe **10**, and if the difference between the first usable time and the second usable time is greater than or equal to the second difference, the drive determination function **3019** drives the cooler **107** with the cooling function of the strong level.

(71) As such, by separating the intensity of the cooling function of the cooler **107**, for instance, if the difference between the first usable time and the second usable time decreases by driving the cooler **107** with the power supplied from the battery **109**, the intensity of the cooling function of the cooler **107** may be gradually weakened. Specifically, if the difference between the first usable time and the second usable time is long, by driving the cooler **107** with the cooling function of strong level and accordingly decreasing the difference of these, the cooler **107** may be driven with the cooling function of medium level and then of weak level. By this, the difference between the first usable time and the second usable time may be made as small as possible.

(72) Likewise, if the difference between the first and the second usable times is small, it is also possible to adjust so that the first usable time becomes a value not too far apart from the second usable time by letting the drive determination function **3019** continuously drive the cooler **107** with the cooling function of weak level.

(73) The cooler **107** having the plurality of different cooling functions may be realized by various cooling methods. For instance, if the cooler **107** is configured by the air-cooling device using a fan etc., the cooling function may be adjusted by modifying an rpm of the fan. Also, for instance, if the cooler **107** is configured by the liquid cooling device where the heat conducting fluid flows inside the pipe, the cooling function may be adjusted by modifying the flowing speed of the heat conducting fluid flowing inside the pipe. Further, for instance, if the cooler **107** is configured by a Piezo element, the cooling function may be adjusted by modifying the voltage applied to the Piezo element.

(74) When driving the cooler **107** with the plurality of different intensities as such, at Step S**19** in the cooling control function of FIG. **7**, the drive determination function **3019** determines what intensity of the cooling function to drive the cooler **107** with when newly driving the inactive cooler **107**, or if the cooler **107** is active, determines whether to modify the intensity of the cooling function or stop the drive.

(75) Then, at Step S**21**, the drive determination function **3019** not only simply drives or stops the cooler **107**, but also newly drives the cooler **107** with a designated intensity of the cooling function or modifies the cooling function of the active cooler **107** to the designated intensity based on the determination of the drive determination function **3019** at Step S**19**.

(76) As described above, according to the ultrasonic diagnostic apparatus **1** of the present embodiment, the usable time of the ultrasonic probe **10** may be extended as possible by making the difference between the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, and the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**, as small as possible. By such, as conventionally, it becomes possible to keep the usable time of the ultrasonic probe **10** extended as possible while evading a performance of the ultrasonic probe **10** decreasing regardless of the operator's intention by automatically switching the ultrasonic probe **10** to a power savings mode.

(77) In other words, by optimizing the relationship between the power consumption of the battery **109** and suppressing the internal temperature rise, it becomes possible to evade a situation of being unable to further use the ultrasonic probe **10** due to the internal temperature rise of the ultrasonic probe **10** even though the battery **109** still has remaining power. Also, in contrast, it becomes possible to evade a situation of being unable to further use the ultrasonic probe **10** due to the remaining level of the battery **109** being empty even though the internal temperature of the ultrasonic probe **10** is still in a usable range.

Modification of the First Embodiment

(78) The ultrasonic probe **10** which the ultrasonic diagnostic apparatus **1** according to the first

embodiment described above includes is based on a premise that the battery **109** is built in the ultrasonic probe **10** and cannot be easily detached. However, it is possible to have the battery **109** detachable from the ultrasonic probe **10** in the ultrasonic diagnostic apparatus **1** described above.
(79) FIG. **8** is a diagram illustrating the exemplary internal layout of the ultrasonic probe **10** where the battery **109** is incorporated to the battery pack **150** and made detachable from the ultrasonic probe **10**, which corresponds to FIG. **3** described above. In a modification shown in FIG. **8**, the battery pack **150** is detachable from the ultrasonic probe **10**, and the operator may charge the battery **109** of the battery pack **150** with a charger by removing the battery pack **150** from the ultrasonic probe **10**.
(80) As shown in FIG. **8**, the battery pack **150** has the cooler **107** and a heat storage **151** in addition to the battery **109**. In other words, in the present modification, the cooler **107** is also incorporated into the battery pack **150** and drive controlled by the processing circuitry **301** in the apparatus body **30** via the processing circuitry **105**. Then, the heat storage **151** is cooled by driving the cooler **107**.
(81) The heat storage **151** has a function of storing heat generated inside the ultrasonic probe **10**. Thus, the usable time of the ultrasonic probe **10** may be extended by suppressing the internal temperature rise of the ultrasonic probe **10**. For instance, the heat storage **151** is configured by a material which transits the phase from solid to liquid and absorbs (stores) latent heat when reaching a temperature higher than a predetermined phase transition temperature and transits the phase from liquid to solid and exhausts (dissipates) latent heat when reaching a temperature lower than the predetermined phase transition temperature. As a specific example, the heat storage **151** may be generated from materials such as organic substances such as wax and paraffin, inorganic hydrates such as sodium sulfate decahydrate and sodium acetate trihydrate, or low melting point metals such as wood metal and gallium.
(82) In the present modification, the cooler **107** cools the heat storage **151**. Then, when the battery pack **150** is removed from the ultrasonic probe **10** and charged, the heat storage **151** is cooled in parallel with the charge. By this, the stored heat of the heat storage **151** may be dissipated in a short amount of time, and the heat storage **151** may be sufficiently cooled when the battery **109** is completely charged.
(83) As such, by arranging the battery **109**, the cooler **107**, and the heat storage **151** as the battery pack **150** detachable from the ultrasonic probe **10**, a continuous usage time of the ultrasonic probe **10** may be extended. Also, the cooler **107** may efficiently suppress the internal temperature rise of the ultrasonic probe **10** by cooling the heat storage **151**.
Second Embodiment
(84) Although the ultrasonic diagnostic apparatus **1** according to the first embodiment described above drive controlled the cooler **107** provided on the ultrasonic probe **10** by the processing circuitry **301** of the apparatus body **30**, the drive control of the cooler **107** may be performed by the processing circuitry **105** of the ultrasonic probe **10**. In a second embodiment, an exemplary configuration where the processing circuitry **105** of the ultrasonic probe **10** controls the drive and stop of the cooler **107** will be described.
(85) FIG. **9** is a block diagram illustrating the exemplary electrical configuration of the ultrasonic diagnostic apparatus **1** according to the second embodiment, which corresponds to FIG. **2** of the first embodiment described above. As shown in FIG. **9**, in the ultrasonic diagnostic apparatus **1** according to the present embodiment, the remaining battery acquiring function **3015**, the first calculation function **3016**, the internal temperature acquiring function **3017**, the second calculation function **3018**, and the drive determination function **3019** described in the first embodiment are provided in the processing circuitry **105** of the ultrasonic probe **10**. For this reason, the processing circuitry **105** of the ultrasonic probe **10** configures the drive controller.
(86) Roles and operations of the remaining battery acquiring function **3015**, the first calculation function **3016**, the internal temperature acquiring function **3017**, the second calculation function **3018**, and the drive determination function **3019** are equivalent to that of the first embodiment

described above. Also, the cooling control process shown in FIG. **7** is executed by the processing circuitry **105** of the ultrasonic probe **10**. However, there is no longer the need to intervene with the communication circuitry **305** and the communication circuitry **111** when the remaining battery acquiring function **3015** acquires the remaining level of the battery **109**, and no longer the need to intervene with the communication circuitry **305** and the communication circuitry **111** when the internal temperature acquiring function **3017** acquires the internal temperature of the ultrasonic probe **10**. In other words, the drive control of the independent cooler **107** is performed by the ultrasonic probe **10**.

(87) As such, even if the function of controlling the drive and stop of the cooler **107** is provided on the ultrasonic probe **10** based on the remaining level of the battery **109** and the internal temperature of the ultrasonic probe **10**, the difference between the first usable time, which is the remaining usage time corresponding to the remaining level of the battery **109**, and the second usable time, which is the remaining usage time corresponding to the internal temperature of the ultrasonic probe **10**, may be made small as possible, extending the usable time of the ultrasonic probe **10** as long as possible.

(88) Note that also in the second embodiment, equivalent to the modification shown in FIG. **8** in the first embodiment described above, the battery **109**, the cooler **107**, and the heat storage **151** may be configured as the battery pack **150** detachable from the ultrasonic probe **10**. Then, by such, the continuous usage time of the ultrasonic probe **10** may be extended.

(89) Note that the word “processor” used in above descriptions means circuits such as, for example, a Central Processing Unit (CPU), a Graphics Processing Unit (GPU), an Application Specific Integrated Circuitry (ASIC), a programmable logic device (for example, a Simple Programmable Logic Apparatus (SPLD), a Complex Programmable Logic Apparatus (CPLD), and a Field Programmable Gate Array (FPGA)). The processor executes functions by reading and executing programs stored in the memory. Note that programs may be configured to be directly integrated in the processor instead of being stored in the memory. In this case, the processor realizes functions by reading and executing programs stored in the circuitry. Note that the processor is not limited to the case arranged as a single processor circuitry, but may be configured as a single processor by combining a plurality of independent circuits to realize functions. Furthermore, a plurality of component elements may be integrated into one processor to realize the functions.

(90) While certain embodiments have been described, these embodiments have been presented by way of example only and are not intended to limit the scope of the inventions. The novel devices and methods described in the present disclosure may be in a variety of other forms. Furthermore, various omissions, substitutions and changes may be made for the embodiments of the devices and method of described in the present disclosure without departing from the spirit of the inventions. The embodiments and their modifications are included in the scope and the subject matter of the invention, and at the same time included in the scope of the claimed inventions and their equivalents.

## Claims

1. An ultrasonic diagnostic apparatus, comprising: an ultrasonic probe configured to be connected to an apparatus body, the ultrasonic probe comprising a battery that supplies power to components of the ultrasonic probe and a cooler that cools inside the ultrasonic probe; and a drive controller configured to drive the cooler based on a remaining level of the battery and an internal temperature of the ultrasonic probe, wherein the ultrasonic probe further comprises a subject contact configured to contact a subject, send ultrasonic waves to the subject, and receive the ultrasonic waves from the subject, and a thermal conductor configured to be connected to conduct heat between the subject contact and the battery; and wherein the cooler cools the thermal conductor.

2. The ultrasonic diagnostic apparatus according to claim 1, wherein the drive controller is further

configured to drive the cooler based on a first usable time and a second usable time, wherein the first usable time is a remaining usage time of the ultrasonic probe corresponding to the remaining level of the battery, and wherein the second usable time is a remaining usage time of the ultrasonic probe corresponding to the internal temperature of the ultrasonic probe.

3. The ultrasonic diagnostic apparatus according to claim 2, wherein the drive controller comprises processing circuitry configured to acquire the remaining level of the battery, calculate the first usable time based on the remaining level of the battery, acquire the internal temperature of the ultrasonic probe, and calculate the second usable time based on the acquired internal temperature.

4. The ultrasonic diagnostic apparatus according to claim 3, wherein the first usable time is a time until the remaining level of the battery drops below a first threshold, and the second usable time is a time until the internal temperature rises above a second threshold.

5. The ultrasonic diagnostic apparatus according to claim 2, wherein the drive controller comprises processing circuitry configured to drive the cooler when the first usable time is longer than the second usable time.

6. The ultrasonic diagnostic apparatus according to claim 5, wherein the processing circuitry is further configured to stop driving the cooler when the first usable time is shorter than the second usable time.

7. The ultrasonic diagnostic apparatus according to claim 2, wherein the drive controller is further configured to control the cooler to decrease a difference between the first usable time and the second usable time.

8. The ultrasonic diagnostic apparatus according to claim 1, wherein the battery is detachable from the ultrasonic probe.

9. The ultrasonic diagnostic apparatus according to claim 8, wherein the battery and a heat storage are configured as a battery pack detachable from the ultrasonic probe.

10. The ultrasonic diagnostic apparatus according to claim 9, wherein the cooler cools the heat storage.

11. The ultrasonic diagnostic apparatus according to claim 10, wherein the battery pack further comprises the cooler.

12. The ultrasonic diagnostic apparatus according to claim 1, wherein the ultrasonic probe is connected to the apparatus body with wireless communication, and wherein the cooler receives power from the battery.

13. An ultrasonic diagnostic apparatus comprising: an ultrasonic probe configured to be connected to an apparatus body, the ultrasonic probe comprising a battery that supplies power to components of the ultrasonic probe and a cooler that cools inside the ultrasonic probe; and a drive controller configured to drive the cooler based on a remaining level of the battery and an internal temperature of the ultrasonic probe, wherein the drive controller is configured to drive the cooler based on a first usable time and a second usable time, wherein the first usable time is a remaining usage time of the ultrasonic probe corresponding to the remaining level of the battery, and wherein the second usable time is a remaining usage time of the ultrasonic probe corresponding to the internal temperature of the ultrasonic probe, wherein the drive controller comprises processing circuitry and the processing circuitry is configured to drive the cooler when the first usable time is longer than the second usable time and to stop driving the cooler when the first usable time is shorter than the second usable time, and wherein the processing circuitry is further configured to drive a cooling function of the cooler with a different intensity in response to how great a difference between the first usable time and the second usable time is.

14. An ultrasonic probe used in an ultrasonic diagnostic apparatus, the ultrasonic probe comprising: a battery configured to supply power to components of the ultrasonic probe; a cooler configured to cool inside the ultrasonic probe; a drive controller configured to drive the cooler based on a remaining level of the battery and an internal temperature of the ultrasonic probe; a subject contact configured to contact a subject, send ultrasonic waves to the subject, and receive the ultrasonic

waves from the subject; and a thermal conductor configured to be connected to conduct heat between the subject contact and the battery, wherein the cooler cools the thermal conductor.
15. The ultrasonic probe according to claim 14, wherein the drive controller is further configured to drive the cooler based on a first usable time and a second usable time, wherein the first usable time is a remaining usage time of the ultrasonic probe corresponding to the remaining level of the battery, and wherein the second usable time is a remaining usage time of the ultrasonic probe corresponding to the internal temperature of the ultrasonic probe.
16. The ultrasonic probe according to claim 15, wherein the drive controller comprises processing circuitry configured to: acquire the remaining level of the battery; calculate the first usable time based on the remaining level of the battery; acquire the internal temperature of the ultrasonic probe; and calculate the second usable time based on the acquired internal temperature.
17. The ultrasonic probe according to claim 16, wherein the first usable time is a time until the remaining level of the battery drops below a first threshold, and the second usable time is a time until the internal temperature rises above a second threshold.
18. The ultrasonic probe according to claim 15, wherein the driver controller comprises processing circuitry configured to drive the cooler when the first usable time is longer than the second usable time.
19. The ultrasonic probe according to claim 18, wherein the processing circuitry is further configured to stop driving the cooler when the first usable time is shorter than the second usable time.